

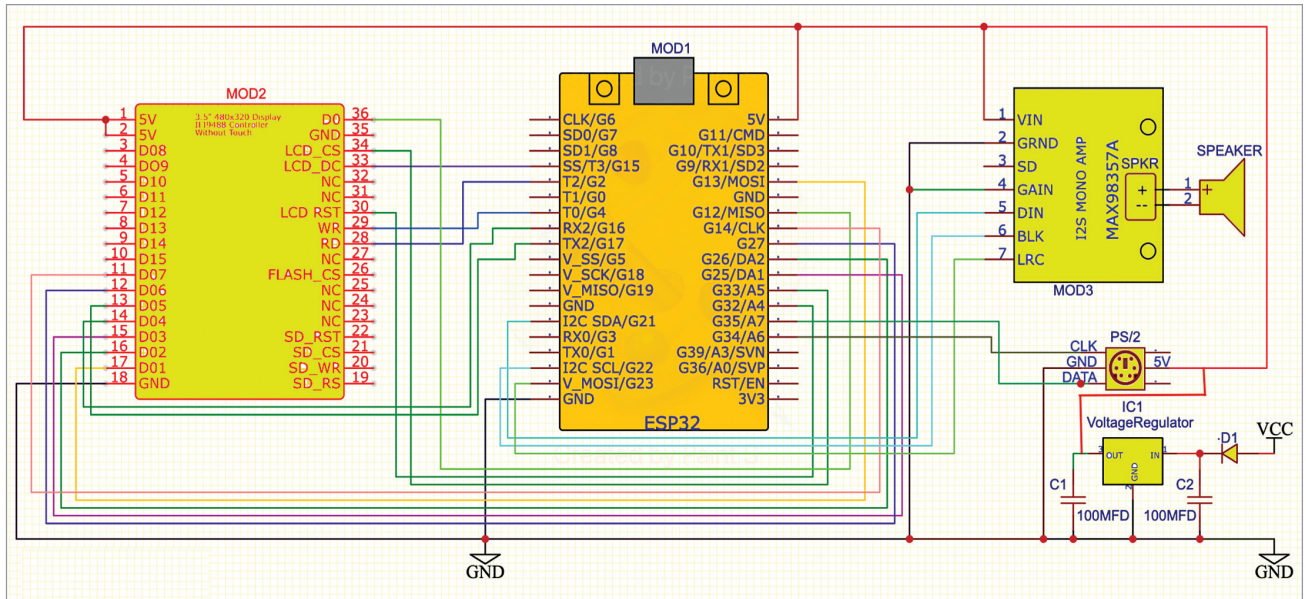


Fig. 2: Circuit diagram

board needs only 4 pins to connect to a microcontroller's 5V, ground, data, clock (or IRQ) pins. Since this is a pure data-in device, some GPIO pins of ESP32 such as G34 and G35 can be connected to it. GPIO pins

G34 and G35 are data-in pins only; they are not data-out pins. Never use them in TFT, LEDs, relays etc, as they need signal out, unlike keyboard pins that only need data in from keyboard. Of course, you can connect a PS2 keyboard to other GPIO pins as well.

Fig. 3 shows the pin details of PS2 keyboard and mouse cable pin details. Fig. 4 show the I2S voice module.

Making ChatGPT talk

Making ChatGPT talk is made simple by the breakthrough with

Google TTS (text-to-speech) and the use of ESP32 board. Here, GTTS is used, because in earlier designs, the few hardware-controlled sound libraries available for free like

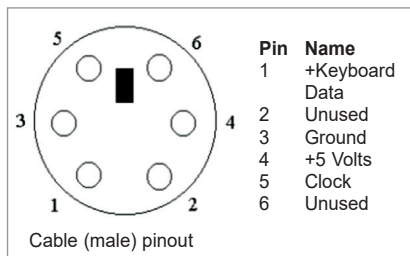


Fig. 3: PS2 keyboard and mouse cable pins

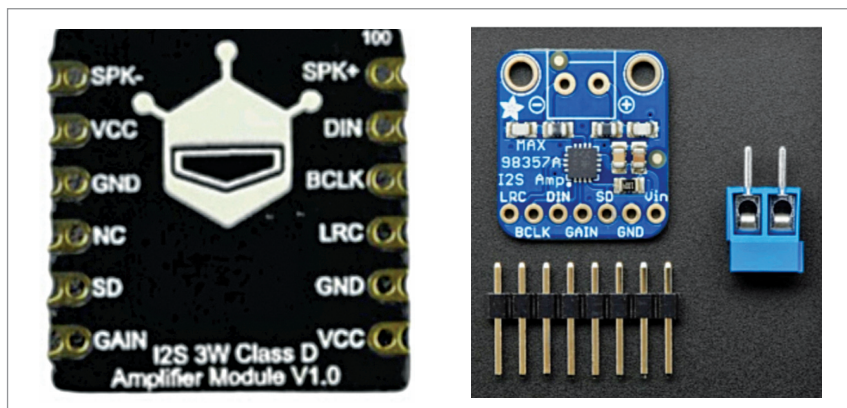


Fig. 4: I2S voice module

TABLE 3
PIN CONNECTIONS
BETWEEN PS2 KEYBOARD
AND ESP32 BOARD

PS2 Keyboard Pin	ESP32 Pin
DATA pin	G35
CLK pin	G34
5V	5V pin of voltage regulator 7805 (IC1)
GND	GND

TABLE 4
PIN CONNECTIONS
BETWEEN MAX98357A
(MOD3) AND ESP32
BOARD (MOD1)

MAX98357A	ESP32 Pin
DIN	G21
BLK	G22
LRC	G23
GND	GND
5V	5V pin of voltage regulator 7805 (IC1)